

Rizwan Shah

AI Researcher | EEG Signal Processing
Deep Learning | Cognitive/Healthcare AI

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Profile

Dedicated **AI Researcher** with a strong background in **Machine Learning, Deep Learning, Computer Vision, and Biomedical/Neuro-Signal Processing**. Experienced in developing data-driven models for EEG-based analysis, biomedical signal processing, healthcare AI, and human-centered computational research. Skilled in working with complex multimodal data, including visual, physiological, and sensor-based signals, to support objective assessment, prediction, classification, and decision-making. Experienced with **scalp EEG data acquisition**, preprocessing, feature extraction, model validation, and research documentation. Committed to advancing interdisciplinary AI research at the intersection of computational modelling, neuroscience, and real-world healthcare applications.

Education

MS Artificial Intelligence	PAF-IAST, Haripur CGPA: 3.58/4.00 ◦ Relevant Coursework: Advanced Machine Learning, Advanced Deep Learning, Advanced Digital Image Processing, Advanced Expert and Recommender Systems, Tools and Techniques for Data Science, Advanced Data Visualization, Computer Vision and Pattern Recognition, Advanced Natural Language Processing.	(2024 – Present)
BS Software Engineering	COMSATS University Islamabad, Abbottabad CGPA: 3.32/4.00 ◦ Relevant Coursework: Object-Oriented Programming, Data Structures, Artificial Intelligence, Software Project Management, Software Requirements Engineering, and Design Patterns.	(2020 – 2024)
DAE Electrical	Govt. College of Technology, Abbottabad Percentage: 74% ◦ Key Subjects: Electrical Machines, Digital Electronics, Circuit Analysis, and Industrial Electronics.	(2017 – 2020)
SSC Science	Oxford Science School, Karak Percentage: 79.8% ◦ Key Subjects: Physics, Chemistry, Mathematics, and Biology.	(2016 – 2017)

Research Interests

- **Biomedical Signal Processing:** Analysis of **EEG, EMG, GSR**, and other physiological signals for healthcare, neuroscience, rehabilitation, and human-state understanding.
- **Computer Vision and Deep Learning:** Development of intelligent vision-based models for image analysis, movement understanding, object detection, and real-world AI applications.
- **Medical Imaging and Healthcare AI:** Machine learning and deep learning approaches for medical image analysis, disease assessment, and clinical decision-support systems.
- **Multimodal Machine Learning:** Fusion of visual, physiological, and sensor-based data for prediction, assessment, classification, and intelligent decision-making.
- **Human-Centered and Assistive AI:** AI-driven systems for rehabilitation, assistive technologies, human behavior analysis, personalized healthcare, and objective functional assessment.

Publications

Rizwan Shah, Amina Riaz, Shahnawaz Qureshi, Ali Zia, and Uzair Iqbal. *"Enhancing Silent Speech Decoding Using EEG and Gradient Boosting Classifiers: A Comparative Analysis of XGBoost and LightGBM on a Native Arabic Silent Speech Dataset."* Sino-Pak Center for Artificial Intelligence, PAF-IAST, Pakistan. DOI: 10.1109/ABC64332.2025.11118350

Research Experience

Research Assistant	PAF-IAST, Haripur (Oct 2024 – Oct 2025)
	- Conducted research in neuro-AI, brain-computer interfacing (BCI), and biomedical signal analysis by integrating EEG signal processing, machine learning, and deep learning techniques. - Designed and executed EEG data acquisition protocols using the Neuroelectronics ENOBIO Research 32 scalp EEG system, supporting participant preparation, signal-quality monitoring, experimental protocol execution, and ethical research practices. - Developed machine learning, deep learning, and ensemble-based models, including Mixture of Experts (MoE), XGBoost, and LightGBM, for multi-band brain signal analysis and classification. - Implemented end-to-end EEG analysis pipelines, including signal preprocessing, artifact handling, feature extraction, model training, and performance evaluation using rigorous validation strategies. - Contributed to technical reports, manuscript preparation, literature review, and research documentation aligned with academic publication standards. - Collaborated with interdisciplinary teams across neuroscience, biomedical engineering, and artificial intelligence to develop clinically relevant and data-driven AI solutions.

Technical Skills

Programming Languages:	Python, Dart, Java, C/C++, C#, JavaScript, SQL
Machine Learning / AI:	TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM
Computer Vision:	OpenCV, YOLO-based models, OCR systems, image preprocessing, object detection
Biomedical Signal Analysis:	Neuroelectronics ENOBIO Research 32, scalp EEG data acquisition, EEG preprocessing, MNE, signal filtering, artifact handling, feature extraction, BCI
Data Science Libraries:	NumPy, Pandas, Matplotlib, Seaborn, Plotly
Quantitative Skills:	experimental protocol design, data cleaning, quantitative evaluation, model validation, research data organization
Development Tools:	VS Code, PyCharm, Android Studio, Google Colab, Jupyter Notebook, Git/GitHub
Databases / Platforms:	Firebase, MongoDB, MySQL, MS Access, SQL Server
Soft Skills:	Research communication, teamwork, leadership, problem solving, time management, mentoring

Professional Experience

Computer Vision Engineer	Ccript Agency (Nov 2025 – Present)
	- Design and implement advanced computer vision pipelines for document understanding, including layout analysis, object detection, and structured information extraction from complex engineering drawings. - Develop and optimize deep learning models using YOLO-based architectures, OCR systems, and vision-language models for scalable processing of high-resolution construction documents. - Build end-to-end AI workflows, including data preprocessing, annotation strategies, model training, fine-tuning, and performance evaluation for real-world deployment. - Improve model robustness through domain-specific feature engineering, augmentation techniques, and iterative error analysis.

- Collaborate with multidisciplinary teams to deliver production-grade AI systems by bridging research methods with practical deployment constraints.

AI/ML Engineer

Firnas.tech

(Nov 2024 – May 2025)

- Contributed to applied AI/ML projects by supporting data preprocessing, model experimentation, evaluation, and documentation.
- Assisted in developing solution-oriented machine learning workflows aligned with organizational and project-specific requirements.
- Collaborated with technical teams in a fast-paced environment, demonstrating ownership, adaptability, and a research-driven problem-solving approach.

Teaching and Leadership Experience

Lab Engineer

PAF-IAST, Haripur

(Sept 2024 – Dec 2025)

- Instructed laboratory sessions for **Programming for AI (Python)**, **Artificial Intelligence**, and **Introduction to Data Science**, covering foundational machine learning concepts such as KNN and linear regression.
- Facilitated hands-on labs using Anaconda, Google Colab, NumPy, Pandas, Matplotlib, and related data science tools.
- Designed and supervised **Digital Image Processing** labs using OpenCV, including thresholding, filtering, histogram equalization, and image enhancement techniques.
- Led **Data Visualization** labs using Matplotlib, Seaborn, and Plotly, and evaluated students through tasks, reports, exams, and viva assessments.
- Maintained lab systems and supported student research, technical troubleshooting, and academic project development.

Trainee Instructor

Microsoft College of Computer Management, Abbottabad (Feb 2024 – Jul 2024)

- Delivered hands-on training in Flutter and Web Development, including HTML, CSS, and JavaScript.
- Designed practical assignments aligned with real-world application development and industry needs.
- Mentored students through personalized guidance to strengthen programming fundamentals and technical confidence.

IT Guru / Tech Lead

Google Developer Student Club, CUI Abbottabad

(Sept 2023 – Jul 2024)

- Led the IT Ninjas team by managing technical infrastructure, platforms, and event-related technical support.
- Provided technical support during the Google Solution Challenge under time-sensitive delivery requirements.

Vice President

AI Community of Pakistan

(Aug 2023 – Jul 2024)

- Directed strategic initiatives to expand community engagement and promote AI learning opportunities.
- Organized workshops, seminars, and hackathons to support AI education, innovation, collaboration, and mentorship.

Awards and Achievements

2025	Recipient, DEI Scholarship – NIRx Medical Technologies
2024	Awarded Fully Funded Master's Scholarship – PAF-IAST
2024	Finalist, Google Global Solution Challenge
2024	1st Place, Build with AI Idea Competition – GDSC, CUI Abbottabad
2024	Runner-Up, SparkPitch Idea Competition – Islamia College
2023	Shield of Resource Person for Flutter Workshop – GCT Abbottabad
2023	Outstanding Performance Award – Google Developer Student Club

Certifications and Trainings

2025	NRG Summer School on AI for Brain-Controlled Applications – Air University, Islamabad
2025	Supervised Machine Learning: Regression – Coursera (IBM)
2025	Exploratory Data Analysis for Machine Learning – Coursera (IBM)
2024	Probability and Probability Distributions – Great Learning
2024	Generative AI Explorer: Vertex AI – Google
2024	Generative AI Fundamentals – Google
2024	Introduction to AI and ML on Google Cloud – Google
2024	Introduction to Generative AI – Google
2024	Introduction to Generative AI Studio – Google
2024	Introduction to Responsible AI – Google
2024	Certificate of Appreciation – AI Community of Pakistan
2024	Web Development Workshop – HTML, CSS, JavaScript
2024	Entrepreneurship Skills Training Program – R.E.D International
2023	Introduction to Flask Python – Great Learning
2023	Introduction to TensorFlow and Keras – Great Learning
2023	Certified Python Elementary Programmer – Python Career Trainers
2023	Flutter Bloc State Management and Firebase – Udemy

References

Dr. Shahnawaz Qureshi	Assistant Professor, PAF-IAST, Pakistan shahnawaz.qureshi@paf-iaast.edu.pk
Dr. Nuhman-ul-Haq	Assistant Professor, CUI Abbottabad, Pakistan nuhman@cuiatd.edu.pk